Does the source of the carrier image affect steganographic detectability?

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IMAGE A
no secret

IMAGE B
hidden message
(0.05 bpp)

IMAGE C
hidden message
(0.30 bpp)

Where do you hide a secret — in a real photograph or in an AI image?

Three pictures of two elephants. To your eye they are identical. To a steganalyst, the **0.05 bpp** payload in Image B touches 2.5% of the LSBs; the **0.30 bpp** in Image C touches 15%. Both are undetectable by inspection but loud to the right test.

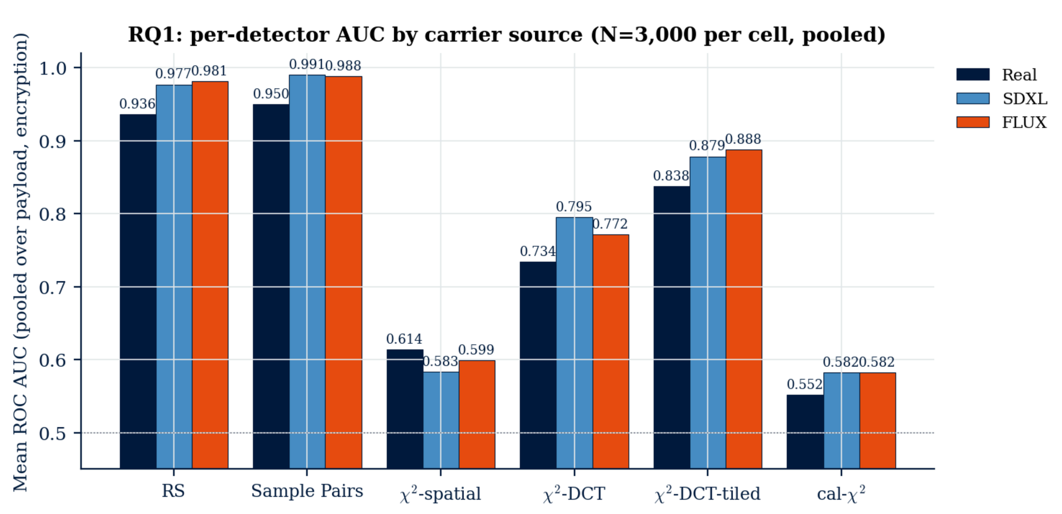
We ran the test on **3,000 caption-matched groups** of real photos and diffusion-generated images, with six classical detectors and two deep learned ones. Three findings ↓

FINDING 1

AI carriers are slightly easier to attack

Five of six training-free detectors find diffusion carriers easier than real photographs. The pooled effect is small ($|\Delta_{AUC}| \approx 0.013$), **3 to 7×** larger in the operationally screened FPR ≤ 0.10 tail.

5/6

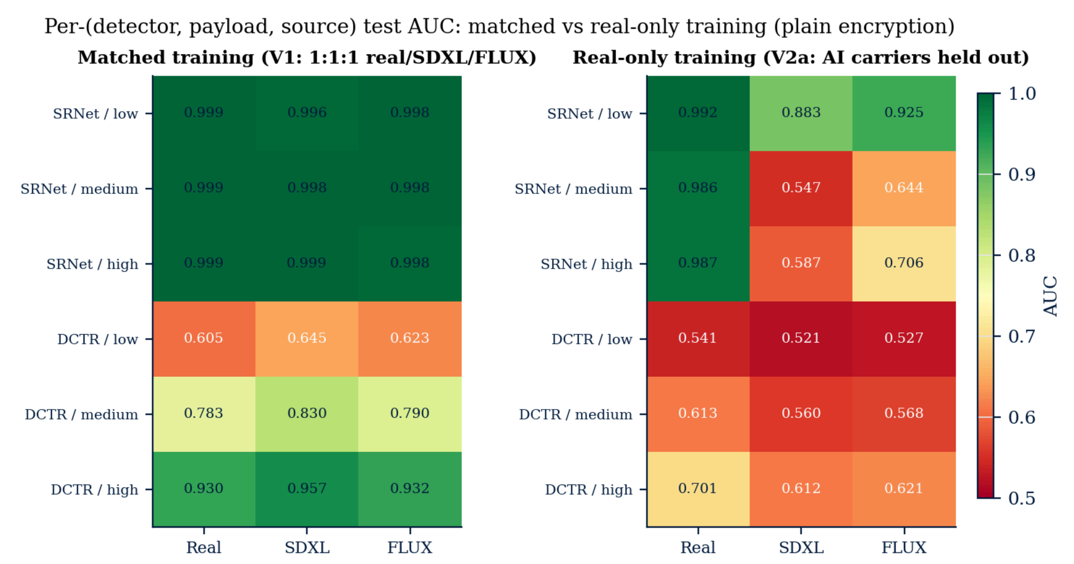


FINDING 2

A deep CNN is fooled by its training set

SRNet trained on a balanced real + AI corpus looks source-invariant. Trained on **real photos only**, it interprets diffusion-decoder noise as the LSB perturbation it was taught to detect. Cover-stego separation collapses on AI images.

+0.44



FINDING 3

A new detector wins on the reference dataset

Our **tile-local χ^2-DCT** detector partitions the DCT-block grid into T×T disjoint tiles and runs Westfeld's pairs-of-values test in each. It beats the textbook global baseline on BOSSBase 1.01 at both quality factors.

+0.09

Q	Embedder	global	tile-local	Δ
75	JSteg	0.764	0.854	+0.090
95	JSteg	0.816	0.902	+0.086
75	OutGuess	0.502	0.500	-0.002
95	OutGuess	0.507	0.507	0.000

METHOD

A pre-registered five-stage pipeline

1 / DATA — **Cover** — real + SDXL + FLUX, caption-matched, 512×512 grayscale →
2 / EMBED — **Stego** — LSB on PNG + JSteg-style DCT-LSB on JPEG Q=95 →
3 / DETECT — **Score** — 6 training-free + 2 learned detectors →
4 / STATS — **Contrast** — Caption-clustered bootstrap + Bonferroni–Holm →
5 / VERDICT — **Decide** — Pre-registered δ_{min} = 0.05 practical-significance gate

3,000 caption groups · 108k stegos · 648k classical predictions · cross-validated on a separate matched training corpus · seed 4242 · **reproducible from the repo**.

Try it yourself

Encode a message into a real photograph or an AI image in your browser. Watch which of the six detectors catch it — and which don't.

→ SCAN TO LAUNCH THE DEMO

/TOOLBOX

SCOPE Non-adaptive LSB embedding only (adaptive variants like J-UNIWARD / HILL / MiPOD deliberately out of scope: the χ^2 family they evade is the whole point of the comparison). Frequency-branch real covers carry a double-JPEG history that AI covers do not — treated as a first-class confound (paper §IX). Real-only training uses Flickr30k only; test set mixes COCO + Flickr30k — magnitude argument holds (effect » cross-dataset shift) but flagged.

Full paper, code, data & reproducible figures

17-page IEEE-TIFS-style paper · complete predictions CSVs · BOSSBase tile-local results · 245 automated tests · reproducible from seed 4242.
gitlab.maastrichtuniversity.nl/m2-2_group02/m2-2_steganography

CODE & DATA